Early Black-Hole Growth Constraints from JWST

Nicole Jiang

Draft: 7 September 2026

Abstract

Aims. Massive black holes in the early universe must grow within limited cosmic time. We identify which JWST black-hole candidates test early growth and which assumptions control their interpretation. **Methods.** We compile 350 literature measurements from 32 source families at $z \geq 4$. Excluding unresolved identities gives 224 primary objects with comparable mass methods and 234 objects in an inclusive exploratory sample. We propagate quoted mass errors, test coherent mass-scale shifts, and compare fixed seed, formation-time, and efficiency assumptions. **Results.** For a $10^2 M_\odot$ seed formed at $z = 30$, radiative efficiency $\epsilon = 0.1$, and no merger boost, 12 primary objects require a lifetime-average Eddington ratio above unity; six retain conditional probability at least 0.95 under their reported errors. UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604 retain this probability threshold after a -0.5 dex mass shift, although source-specific lensing, scattering, and calibration assumptions remain relevant. The most demanding object changes from UNCOVER-20466 to the exploratory object GN-z11 when seeding is delayed to $z = 20$. A separate A2744-QSO1 dynamical-mass comparison moves its requirement close to unity, with an error interval crossing the threshold. **Conclusions.** Using the catalogued masses and redshifts, we determine which seed masses and radiative efficiencies permit growth to the observed masses within the available cosmic time. Lowering the efficiency to $\epsilon = 1 - \sqrt{8/9} \simeq 0.05719$, the ideal thin-disc value for a non-spinning (Schwarzschild) black hole, permits all central mass estimates in the primary and inclusive exploratory samples to be reached from $10^2 M_\odot$ seeds formed at $z = 30$ without requiring lifetime-average accretion above the Eddington limit.

Key words. black hole physics – accretion, accretion disks – galaxies: active – galaxies: high-redshift – quasars: supermassive black holes

1 Introduction

The discovery of massive black holes in the early Universe raises a fundamental question: what growth histories could have assembled them within the available cosmic time? The strength of this constraint depends on both the observed black-hole mass and redshift, as well as assumptions about seed formation, radiative efficiency, and the duration of active accretion. An apparently demanding object under one set of assumptions may therefore be compatible with less extreme growth under another.

Here, we use a literature-based catalogue of JWST-identified high-redshift accreting black holes and candidates to ask which objects meaningfully constrain early growth histories. The atlas covers $z \geq 4$, extending from the first billion years to a cosmic age of approximately 1.54 Gyr at its lower-redshift boundary. We calculate the average accretion rates required across a range of seed and growth assumptions, incorporating reported mass uncertainties and distinguishing measurements with different evidential and methodological limitations. Source-native measurements, identity links, and mass provenance are retained to support comparison across heterogeneous literature samples.

Our aim is to identify which growth requirements remain demanding across plausible scenarios, which depend primarily on uncertain assumptions, and where improved observations would most strengthen the constraints. This approach treats the catalogue as a means of testing possible growth histories, rather than assuming that every high-redshift black hole presents the same assembly problem. The constraints apply to individual objects under stated assumptions.

Analytic seed-growth comparisons already establish the trade-off between seed mass and sustained accretion. For example, Dayal [4] considered light and heavy seeds at $z_{\rm seed} = 25$ and explored a primordial-seeding interpretation. Here we use this established growth framework to assess how object-level constraints change when published masses, measurement methods, and growth assumptions are treated consistently across a larger evidence atlas. The added scientific product is an object-by-object distinction between requirements that survive quoted errors, those that survive a mass-scale perturbation, and those whose interpretation depends on a specific estimator. We report the complete primary threshold set with source-linked caveats and proposed observations, isolate measurement and starting-time effects in a matched literature comparison, and evaluate an independent dynamical estimate outside the frozen catalogue. These comparisons make the catalogue useful for selecting observational tests of growth assumptions. Survey studies provide a complementary perspective: the JADES broad-line census [5] measures current accretion and host scaling relations, while the narrow-line census [24] broadens the evidence for obscured activity. Our comparison asks what cumulative growth the adopted masses require, which is distinct from measuring present accretion or counting AGN in a survey. It uses those source measurements without imposing a common selection function. The source-family inventory is given in Appendix A.

1.1 Theoretical framework

We use the luminosity-based Eddington ratio $f_{\rm Edd} = L_{\rm bol}/L_{\rm Edd}$ and a fixed radiative efficiency ϵ, the fraction of accreted rest-mass energy emitted as radiation. For the available growth interval,

$$\Delta t = t(z_{\rm obs}) - t(z_{\rm seed}) > 0, \qquad \overline{f_{\rm Edd}} = \frac{1}{\Delta t}\int_{t(z_{\rm seed})}^{t(z_{\rm obs})} f_{\rm Edd}(t)\,dt. \tag{1}$$

The analytic growth relation adopted by Dayal [4] gives

$$M_{\rm BH}(t_{\rm obs}) = B_{\rm merge} M_{\rm seed} \exp\left[\frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}}\right], \tag{2}$$

where $t_{\rm Edd} \simeq 0.45$ Gyr. The merger multiplier is $B_{\rm merge} = 1$ for the reference calculation and 2 for a factor-of-two mass boost. This prescription rescales the seed normalization; it does not simulate merger times, mass ratios, gravitational-wave losses or a merger tree. The required lifetime-average ratio is

$$\overline{f_{\rm Edd}}_{\rm req} = \max\left[0, \frac{\epsilon}{1-\epsilon}\frac{t_{\rm Edd}}{\Delta t}\ln\left(\frac{M_{\rm BH}}{B_{\rm merge}M_{\rm seed}}\right)\right]. \tag{3}$$

If the boosted seed already exceeds the observed mass, the requirement is set to zero; this does not make the specified history an exact mass match. A lifetime-average requirement is distinct from an instantaneous Eddington ratio and does not uniquely determine a duty cycle.

Cosmic ages assume a flat matter-plus-Lambda cosmology with $H_0 = 67.3\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$ and $\Omega_\Lambda = 0.685$, neglecting radiation. Appendix D gives the supporting relations, implementation and inverse seed-mass diagnostic. The fixed-efficiency assumption is essential to Eqs. 2–3. The supplementary spin-dependent compatibility prescription is specified in Appendix C; it does not model spin evolution.

2 Catalogue construction

We compile 350 literature measurements from 32 source families at $z \geq 4$. Each adopted mass is linked to its source, method and reported uncertainty. One preferred measurement is selected per object using documented source assessments; alternate measurements remain available for sensitivity tests and are not averaged into the preferred value.

Growth analysis requires a numerical mass and redshift, an identified mass method and a resolved lensing treatment. Upper limits, indirect model ranges and masses inferred by assuming an Eddington ratio are excluded from numerical inference. We also exclude all six records associated with three unresolved identity groups, including three records with eligible masses. This leaves 224 primary objects and an inclusive exploratory sample of 234 objects.

The primary sample uses Balmer-line single-epoch virial masses with secure or probable evidence and no conditional broad-line-region interpretation. The ten additional exploratory objects comprise nine candidates and GN-z11, whose UV mass estimator lies outside the primary method group. Missing mass errors do not exclude a point estimate, but prevent probability inference. These cuts improve method comparability without implying a common calibration or survey selection function. Appendix A gives the source inventory, identity decisions and sample accounting (Figure 6).

3 Growth-model comparison

3.1 Reference model

We calculate the required lifetime-average Eddington ratio using Eq. 3, with $M_{\rm seed} = 10^2 M_\odot$, $z_{\rm seed} = 30$, $\epsilon = 0.1$ and $B_{\rm merge} = 1$ as the reference history. We refer to this conditional requirement as *growth pressure*. Figure 1 illustrates tracks at four fixed efficiencies; tracks are selected for proximity to the observed masses, not fitted growth histories. Appendix D.5 specifies the display selection.

3.2 Uncertainty and sensitivity

We propagate reported asymmetric log-mass errors through 10,000 draws per object, assigning half the draws to each side of the central mass with the corresponding half-normal scale. Redshifts are held fixed. This approximates quoted intervals rather than reconstructing source posteriors. The 12 primary objects without reported errors contribute point estimates only; probability calculations use 212 primary and 222 inclusive exploratory objects. No additional calibration scatter is added to the quoted errors.

We test mass-scale sensitivity by shifting all adopted log masses and their error draws by offsets of -0.5, -0.3, 0, $+0.3$ and $+0.5$ dex, keeping the sample and growth assumptions fixed. These offsets probe the single-epoch mass uncertainty scale [1]; they are stress tests, not inferred calibration corrections. Probabilities remain conditional on each offset. Sampling and conversion details are given in Appendix D.5.

3.3 Compatibility across growth assumptions

A fixed history is compatible with an object when its required seed mass (Eq. 15) lies within the chosen interval in $\log_{10}(M_{\rm seed}/M_\odot)$: $[1, 2]$ (light), $[3, 4]$ (intermediate), $[4, 6]$ (heavy), or $[2, 6]$ (broad). The grid uses $z_{\rm seed} = 30$, three spin/efficiency cases, merger multipliers 1 and 2, and average rates 0.3, 1 and 2. Unlike the reference ranking and fixed-efficiency growth tracks, these maps use the supercritical efficiency prescription in Appendix C.

Seed intervals overlap and carry no probability weights; the broad interval has no formation-channel interpretation. A required seed below an interval means that the specified history overgrows the observed mass. Compatibility fractions describe the admitted objects, not population frequencies, because the literature sample has no common selection function.

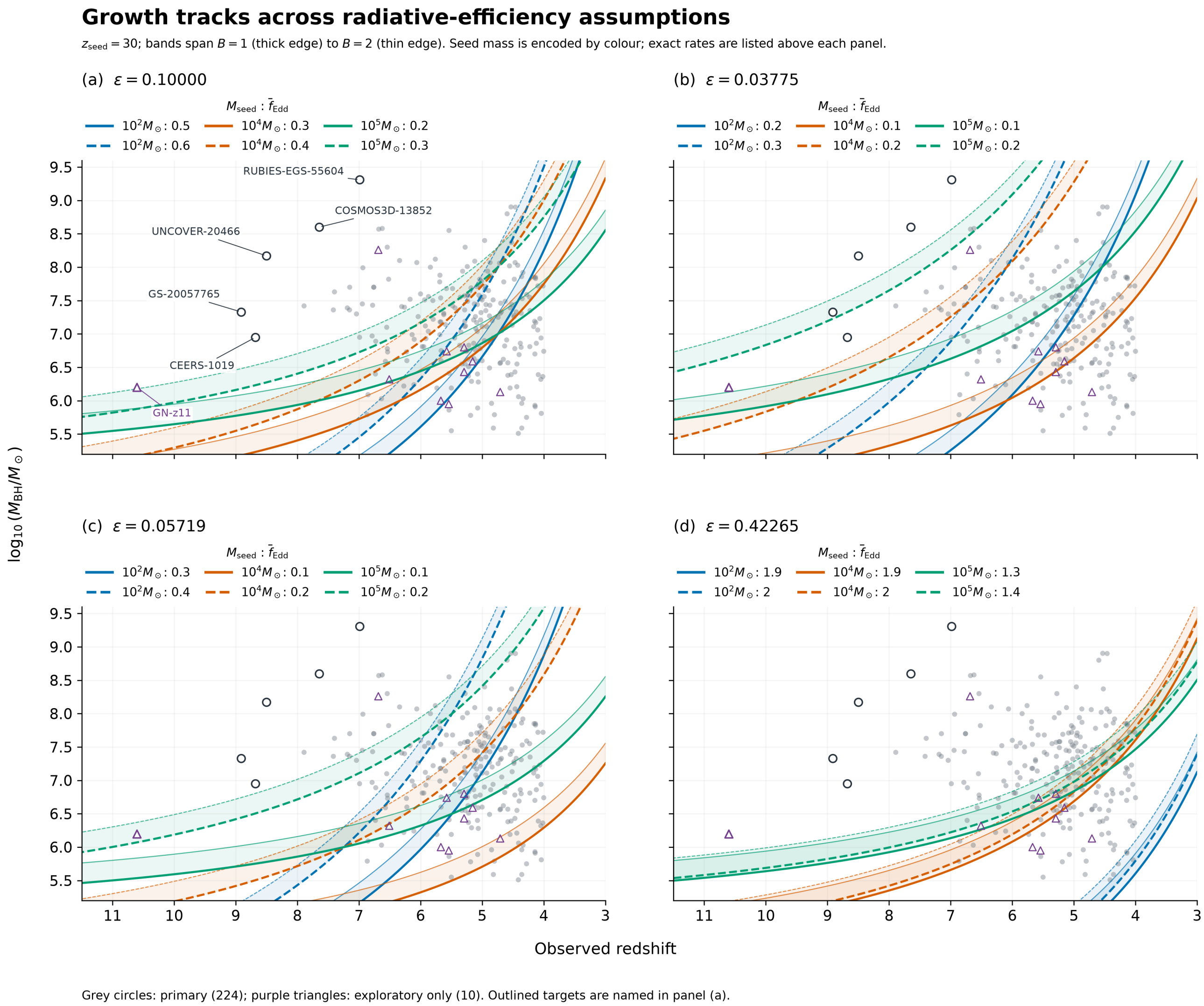

Figure 1: Growth tracks with 224 primary and ten exploratory-only objects. Panels show four constant radiative efficiencies at $z_{\mathrm{seed}} = 30$; colours distinguish seeds of 10^2, 10^4 and $10^5 M_\odot$, and legends list the selected average Eddington ratios. Shaded bands span merger boosts $B = 1$–2, with thick and thin edges respectively; they are not uncertainty intervals. Six named objects are outlined in every panel. The reversed redshift axis spans $z = 11.5$ to $z = 3$. Excluded identities and objects without eligible masses are omitted.

4 Results

4.1 Growth pressure under the reference assumptions

In the 224-object manuscript primary sample, the required f_{Edd} distribution for a $10^2\,M_\odot$ seed has median 0.578 and interquartile range 0.487–0.682. Twelve objects require $f_{\mathrm{Edd}} > 1$ at their point estimates; none requires $f_{\mathrm{Edd}} > 2$. The manuscript exploratory 234-object sample has median 0.574, interquartile range 0.488–0.679, and 14 point requirements above unity. The high-pressure tail is not simply a mass ranking because less cosmic time can offset a lower observed mass. GN-z11, for example, has a lower standardized mass than many broad-line objects but has the second-largest required f_{Edd} at $z = 10.603$. Its UV virial estimator places it outside the primary comparison subset; this position is a conditional cross-method diagnostic, not a claim of equivalent mass-estimator reliability.

UNCOVER-20466 has the largest required average rate in both samples. Its point-estimate required $f_{\rm Edd}$ is 1.452, and its Monte Carlo median is 1.453 with a 16th–84th percentile interval of 1.354–1.549. Table 1 reports all 12 primary point requirements above unity, including the six that meet the conditional probability threshold. Ordering by point requirement differs from ordering by quoted-error probability.

Table 1: Complete primary reference-threshold set, ordered by point requirement. Masses and errors are in $\log_{10}(M/M_\odot)$; all estimators are single-epoch virial. f_0 is the reference point requirement, f_{16} its reported-error 16th percentile, and $P_0 = P(\overline{f_{\rm Edd}}_{\rm req} > 1)$. f_- and P_- apply the fixed -0.5 dex offset. Rounded probabilities above 0.999 do not imply physical certainty. Source-specific caveats and proposed observations are in Table 6.

Object	Line/source	Log mass	f_0	f_{16}	P_0	f_-	P_-
UNCOVER-20466	Hβ [13]	$8.17^{+0.42}_{-0.42}$	1.452	1.354	> 0.999	1.335	> 0.999
GS-20057765	Hβ [5]	$7.33^{+0.62}_{-0.70}$	1.355	1.177	0.980	1.228	0.901
COSMOS3D-13852	Hβ [38]	$8.60^{+0.35}_{-0.33}$	1.314	1.247	> 0.999	1.214	> 0.999
RUBIES-EGS-55604	Hβ [14]	$9.31^{+0.10}_{-0.10}$	1.267	1.250	> 0.999	1.180	> 0.999
CEERS-1019	Hβ [16]	$6.95^{+0.37}_{-0.37}$	1.205	1.115	0.988	1.083	0.822
GS-20030333	Hβ [5]	$7.42^{+0.65}_{-0.48}$	1.133	1.033	0.907	1.029	0.613
GS-164055	Hβ [5]	$7.63^{+0.59}_{-0.66}$	1.065	0.941	0.696	0.970	0.395
GN-38509	Hα [5]	$8.57^{+0.37}_{-0.38}$	1.065	1.005	0.860	0.984	0.392
J0910_2028_12910	Hα [19]	$8.58^{+0.01}_{-0.01}$	1.053	1.051	> 0.999	0.973	< 0.001
CEERS-00717	Hα [9]	$7.99^{+0.16}_{-0.17}$	1.027	0.998	0.821	0.942	0.018
ZS7	Hβ [18]	$7.70^{+0.40}_{-0.40}$	1.023	0.954	0.625	0.934	0.173
GN-4685	Hβ [5]	$7.36^{+0.45}_{-0.42}$	1.017	0.938	0.585	0.922	0.181

4.2 Robustness to reported mass uncertainty

Within the primary sample, eight objects remain above unity at the 16th percentile and six have $P(f_{\rm Edd} > 1) \geq 0.95$, compared with 12 above unity at their point estimates. In the exploratory sample, 14 objects have median required $f_{\rm Edd} > 1$, ten remain above unity at the 16th percentile, and 15 cross unity by the 84th percentile; eight have $P(f_{\rm Edd} > 1) \geq 0.95$. These probabilities use the reported-error model in Section 3.2 and the fixed reference growth assumptions. Figure 2 resolves the primary threshold objects by name. The remaining sample includes 12 primary objects without quoted mass errors; they enter point counts but have no assigned probability. The full sample has 212 primary and 222 exploratory objects with reported errors.

4.3 Sensitivity to the virial-mass scale

Table 2 shows that the exact reference counts are sensitive to the adopted mass scale. A downward shift of 0.3 dex changes the primary 12/8/6 counts to 9/6/4; a downward shift of 0.5 dex leaves 6/4/3. Positive shifts increase all three counts. Thus some reference-history tension survives even the negative 0.5-dex stress test, but the size of that tail is not robust to mass calibration. This does not remove the efficiency degeneracy: the lower-efficiency scenario below is a separate change to the growth model. The three primary objects retaining $P \geq 0.95$ at -0.5 dex are UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604 (Table 1). This selects candidates for estimator tests rather than three established super-Eddington growth histories. In particular, a larger directional scattering bias remains possible for COSMOS3D-13852, as discussed below.

We jointly test publication-version sensitivity and the manuscript identity exclusions by replacing redshifts, masses, and quoted errors for 44 exact-ID, position-confirmed Baccus counterparts with the published Table 1 values [19]. Of five unmatched frozen records, one is already excluded by the identity policy; the remaining four are either retained or omitted. All six identity exclusions remain in force, and the

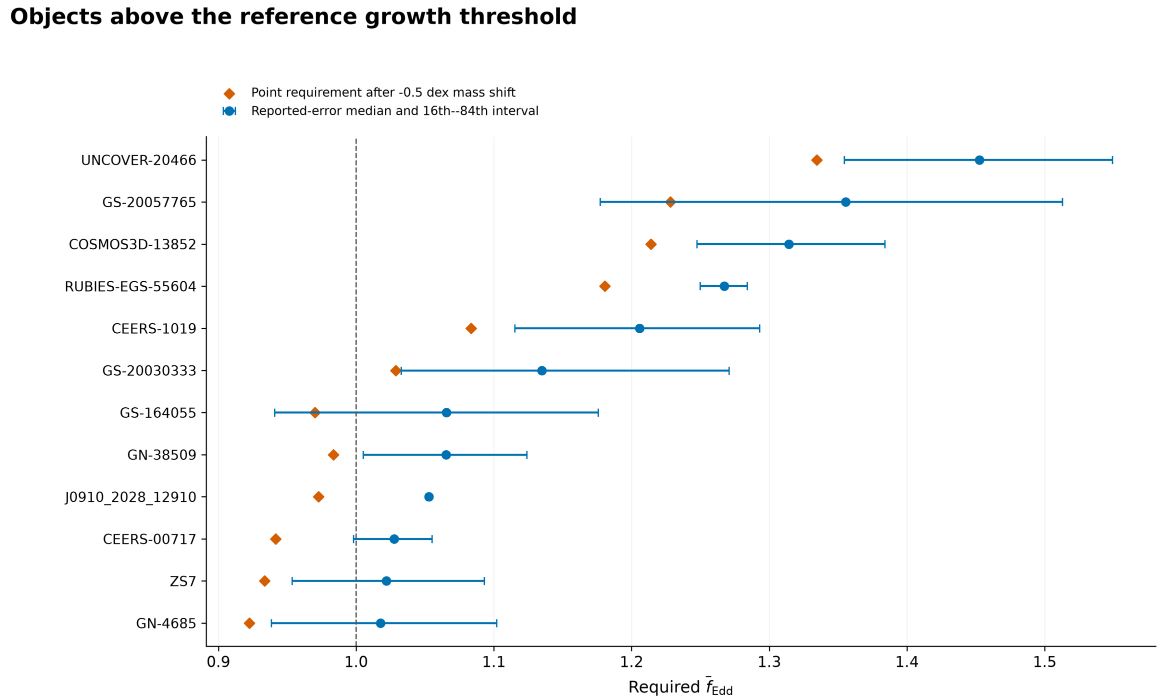

Figure 2: The 12 primary objects whose reference point requirements exceed unity. Circles and bars show reported-error medians and 16th–84th percentile intervals; diamonds show point requirements after a −0.5 dex mass shift. The vertical line is the reference threshold. The selected objects all have reported errors; these intervals do not include additional estimator systematics.

Table 2: Fixed mass-scale stress tests under the reference growth history. Triplets give counts above unity at the point estimate, at the 16th percentile, and with conditional $P(\overline{f_{\mathrm{Edd}}}_{\mathrm{req}} > 1) \geq 0.95$. The probability-based counts use 212 primary and 222 exploratory objects with reported errors; point counts use all 224 and 234 objects.

Mass offset (dex)	Primary	Exploratory
−0.5	6 / 4 / 3	7 / 5 / 4
−0.3	9 / 6 / 4	10 / 7 / 5
0	12 / 8 / 6	14 / 10 / 8
+0.3	15 / 13 / 12	17 / 15 / 14
+0.5	19 / 16 / 13	21 / 18 / 15

evidence/method taxonomy and growth parameters are fixed. We recompute reported-error probabilities with the same identifier-seeded sampling scheme.

Both treatments preserve the primary 12/8/6 and exploratory 14/10/8 threshold counts and each sample's top-five ordering (Table 3). Omitting unmatched records reduces the samples to 220 and 230 objects. The 32 changed central masses shift by −0.10 to +0.04 dex. This combined check supports the stated threshold results under the tested publication update; it does not admit the complete published sample or resolve the excluded identities. Object-level comparisons accompany the analysis.

Table 3: Joint Baccus publication-update and identity-exclusion test. Every row excludes all open identity groups. Counts denote point requirements above unity, 16th percentiles above unity, and conditional $P \geq 0.95$, respectively. Frozen values define the manuscript baseline.

Sample	Measurement treatment	Objects	Counts
Primary	Frozen values	224	12/8/6
Primary	Published; retain unmatched	224	12/8/6
Primary	Published; omit unmatched	220	12/8/6
Exploratory	Frozen values	234	14/10/8
Exploratory	Published; retain unmatched	234	14/10/8
Exploratory	Published; omit unmatched	230	14/10/8

4.4 Model compatibility and measurement choice

Table 4 isolates changes to the reference history using the same preferred masses. Increasing the seed mass from 10^2 to $10^3\,M_\odot$ reduces the primary count above unity from 12 to four; a $10^5\,M_\odot$ seed removes all point requirements above unity. Delaying seed formation from $z = 30$ to $z = 20$ instead raises that count to 22. Lowering the efficiency to 0.05719 removes all point requirements above unity in both samples. These calculations change one assumption at a time; they are sensitivity benchmarks, not a joint distribution of physically realized histories. The seed endpoints bracket the light/heavy alternatives considered in analytic work [4]; $10^3\,M_\odot$ probes the transition between them. The $z = 20$ case tests a shorter available growth interval, and the lower efficiency is the ideal non-spinning thin-disk limit from Eq. 7. None of these choices is assigned a population weight or inferred from the data.

Table 4: Point-estimate requirements above $\overline{f_{\rm Edd}} = 1$ under fixed histories. Each row changes only the named parameter from the reference ($M_{\rm seed} = 10^2 M_\odot$, $z_{\rm seed} = 30$, $\epsilon = 0.1$, $B_{\rm merge} = 1$). Counts use provisional object identities and do not include uncertainty tails.

Scenario	Primary (224)	Exploratory (234)
Reference	12	14
$M_{\rm seed} = 10^3 M_\odot$	4	5
$M_{\rm seed} = 10^5 M_\odot$	0	0
$z_{\rm seed} = 20$	22	24
$\epsilon = 1 - \sqrt{8/9}$	0	0

An explicit inclusion/exclusion sensitivity calculation compares these samples with the original 227/237-object sets. Excluding the six ambiguous records preserves the threshold counts and top-five ordering in every scenario in Table 4, as well as the reference 16th-percentile and $P \geq 0.95$ counts. The primary reference median changes from 0.574 to 0.578. Thus the reported early-growth tail does not depend on retaining the unresolved groups, although catalogue-wide uniqueness remains unsettled.

Figure 3 expresses the degeneracy for five named targets: for each seed mass it shows the largest fixed efficiency allowing $\overline{f_{\rm Edd}}_{\rm req} \leq 1$. The curves solve Eq. 3 with $B_{\rm merge} = 1$ and the adopted point masses; efficiencies below each curve permit the threshold, but do not guarantee a sustainable fuel supply. Comparing $z_{\rm seed} = 30$ and 20 exposes the greater timing sensitivity of GN-z11. Figure 4 complements these object-level boundaries with a comparison across the full manuscript samples.

Figure 4 shows how the fraction compatible with each fixed history changes across seed intervals, spin-dependent efficiencies, merger boosts, and accretion rates. Primary and inclusive exploratory samples show similar patterns. For example, in the heavy-seed interval at $a = +1$ and $f = 1$, increasing the merger multiplier from one to two changes the primary compatible fraction from 57 to 70 per cent (59 to

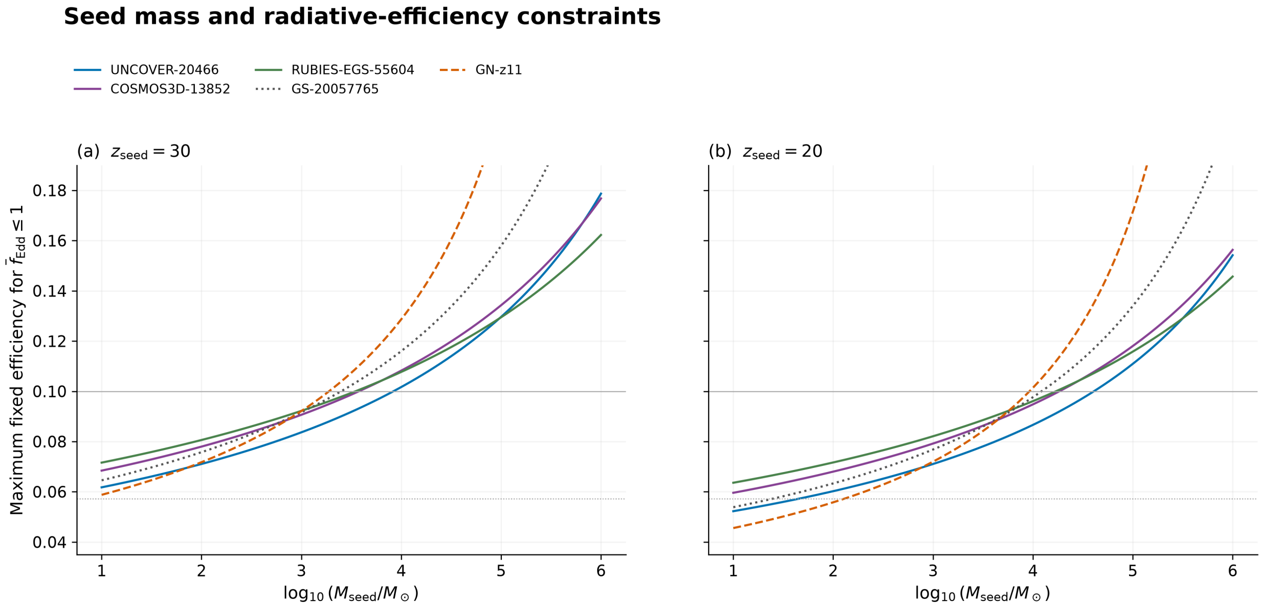

Figure 3: Seed-mass–efficiency boundaries for Eddington-limited average growth at two starting redshifts. Each curve uses an adopted point mass and no merger boost. Below a curve the required average is at most unity. Horizontal guides mark $\epsilon = 0.1$ (solid grey) and $1 - \sqrt{8/9}$ (dotted grey). GN-z11 (dashed) is exploratory; the other objects are primary. These are fixed-efficiency comparisons without probability weights.

71 per cent in the exploratory sample). Compatibility need not increase monotonically with rate: a fixed history can overgrow an object and put its required seed below the interval. These fractions describe point-mass compatibility within the selected samples; they do not include mass-error integration or measure formation-channel frequencies. The efficiency prescription differs from the fixed-$\epsilon = 0.1$ reference ranking and is detailed in Appendix C.

Figure 5 shows seven alternate-measurement comparisons across six objects; CEERS-2782 contributes two alternatives. Alternate-minus- preferred mass shifts range from -0.600 to $+0.949$ dex, while the corresponding required-$f_{\rm Edd}$ shifts range from -0.083 to $+0.121$. All seven comparisons stay below unity at both point estimates. Thus these retained alternatives do not change reference threshold membership for the six tested objects, but they do not establish robustness for the untested high-pressure tail. The follow-up matrix distinguishes reference high-pressure objects, measurement-sensitive cases, and records still requiring a suitable mass.

4.5 Separating measurement and starting-time changes

Table 5 isolates two effects for four unambiguous named matches to Dayal's Table 1 [4]. All columns use our cosmology, $\epsilon = 0.1$, $M_{\rm seed} = 10^2 M_\odot$, and $B_{\rm merge} = 1$. The first uses that table's mass and redshift at $z_{\rm seed} = 25$; the second changes only to our adopted mass and redshift; the third changes only the starting redshift to 30. Thus this is a controlled input comparison, not an exact reproduction of the earlier paper or a complete crossmatch. For these examples the earlier starting time lowers the requirement, whereas measurement updates have smaller effects.

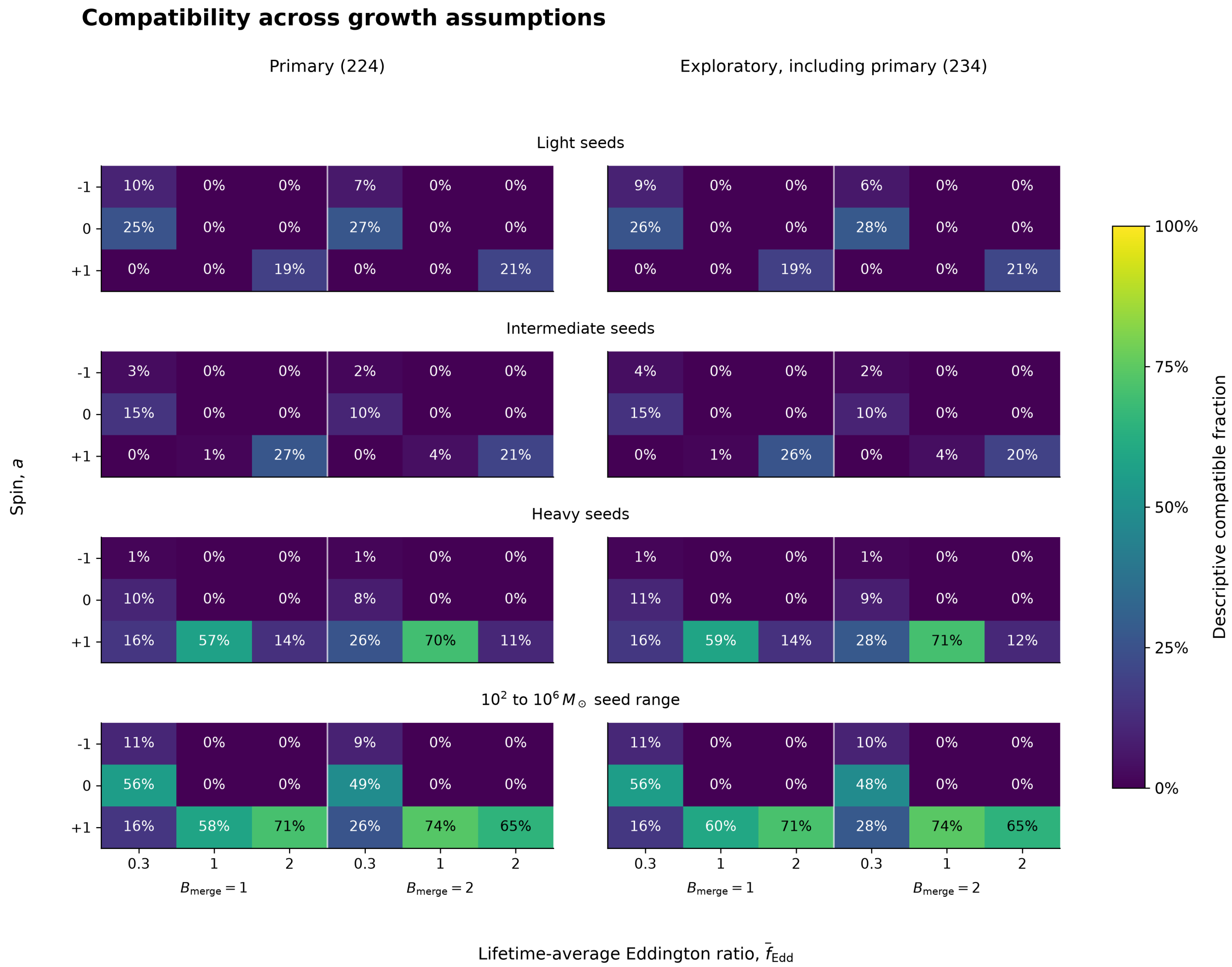

Figure 4: Constant-state compatibility fractions in the primary sample (left) and the inclusive exploratory sample (right). These samples are nested. Each column specifies merger multiplier B and fixed luminosity ratio f; rows specify ideal spin-dependent efficiency, with the supercritical prescription in Appendix C. Seed intervals overlap and carry no probability weights. The broad mass interval has no formation-channel interpretation.

Table 5: Controlled comparison for four named literature matches. Columns give required average $f_{\rm Edd}$; the first two share $z_{\rm seed} = 25$. The external inputs and source locators are released with the analysis.

Object	Earlier inputs, 25	Adopted inputs, 25	Adopted inputs, 30
GN-z11	1.577	1.578	1.437
A2744-QSO1	0.974	0.974	0.929
UNCOVER-20466	1.540	1.548	1.452
CEERS-00717	1.074	1.076	1.027

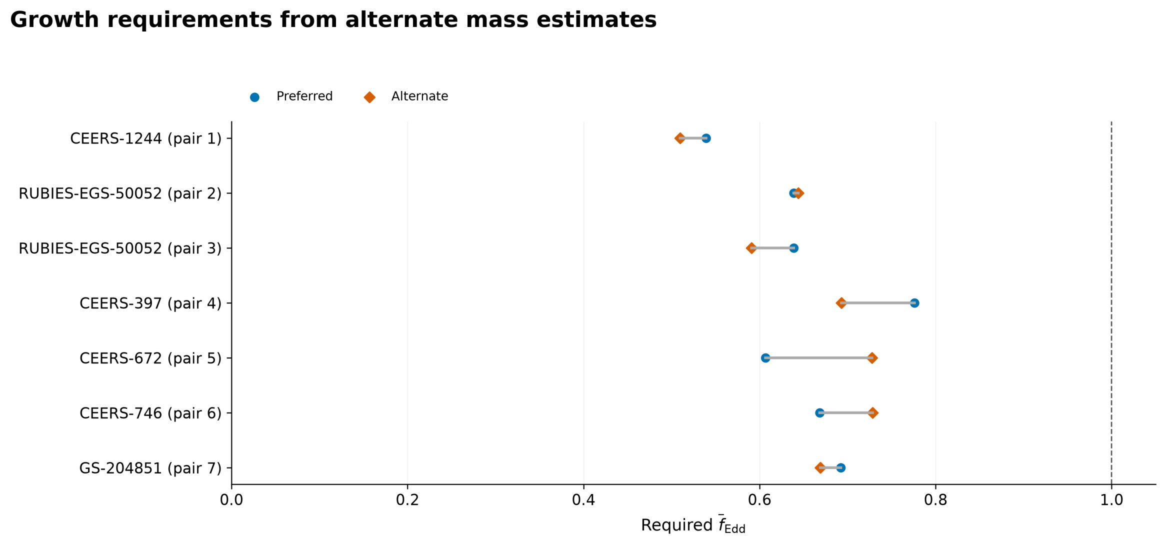

Figure 5: Preferred (circles) and alternate (diamonds) point requirements for seven measurement pairs across six primary objects. Pair numbers follow the released alternate-measurement table, which supplies stable measurement IDs; RUBIES-EGS-50052 (CEERS-2782) has two alternatives. All pairs remain below the reference threshold.

4.6 External dynamical-mass check: A2744-QSO1

The frozen primary catalogue adopts $\log_{10}(M/M_\odot) = 7.3 \pm 0.2$ for A2744-QSO1 from Greene et al. [13]. We separately evaluate the MOKA3D point-mass estimate 7.7 ± 0.3 in the pinned preprint [40]. This is an external sensitivity comparison; it neither replaces the preferred row nor changes primary membership. Both estimates use the adopted $z = 7.04$ and the reference growth assumptions. The direct estimate is conditional on the lens reconstruction and fitted inclination; its inclination-free lower-limit counterpart is not sampled as a symmetric measurement.

The point requirement rises from 0.929 to 1.000. With a symmetric normal approximation to the quoted log-mass errors, the exact 16th–84th percentile intervals are 0.895–0.964 and 0.947–1.052, respectively. The direct estimate lies just below unity at the point value, but its interval crosses the threshold; the conditional exceedance probability is 0.497. Rounding the point requirement to 1.000 does not indicate exact equality. The comparison supplies independent dynamical evidence relevant to this object's virial interpretation, and moves it from a sub-threshold reference case to one whose quoted-error interval straddles unity. Agreement for this object does not calibrate the full primary sample. A future dataset may admit the direct measurement with its own method classification and identity link.

5 Discussion

5.1 What the high-pressure tail establishes

The reference requirements above unity are concentrated at $z = 6.6214$–10.603. The principal growth constraint therefore comes from the early tail of the catalogue, while the lower-redshift majority supplies comparison objects.

The reference calculation identifies a small set of objects that are difficult to grow from $10^2\,M_\odot$ seeds at $f_{\rm Edd} \leq 1$ under the reference $\epsilon = 0.1$, $z_{\rm seed} = 30$ history. The result does not select a unique remedy. Earlier or heavier seeds, lower effective radiative efficiency, merger growth, or episodes above the reference accretion rate can each reduce the tension. The atlas is designed to expose these trade-offs for each object rather than collapse them into a single formation label.

This dependence can be quantified without changing the catalogue. At fixed seed mass, growth interval and merger factor, required $f_{\rm Edd}$ scales as $\epsilon/(1-\epsilon)$. The implemented ideal non-spinning thin-disk value $\epsilon = 1-\sqrt{8/9} = 0.05719$ reduces every point requirement by a factor 0.54594 relative to $\epsilon = 0.1$: the largest becomes 0.793 and none exceeds unity. This sensitivity calculation does not infer actual spins or growth histories; it shows why the reference tail cannot establish a model-independent need for super-Eddington accretion.

The matched comparison in Table 5 separates revised inputs from available growth time. The broader atlas adds a different observational question: which of the reference-threshold objects should be tested first, and for which mass assumption? Table 6 answers this question without treating a high probability under a narrow formal error as an independent validation of the estimator.

The most useful targets are therefore those whose placement is both demanding and stable to published mass uncertainty. Independent mass measurements, better separation of broad-line-region kinematics from scattering or outflows, and constraints on obscuration and lensing can change the physical interpretation more than another evaluation of the same analytic formula. The follow-up matrix makes that distinction explicit, but its heuristic ordering is not a calibrated calculation of expected information gain.

5.2 Estimator credibility and observational tests

Survival of the -0.5 dex test is meaningful only relative to that perturbation. UNCOVER-20466 uses a demagnified Hβ mass and still requires checks of the lens model and broad-line interpretation [13]. RUBIES-EGS-55604 uses a dust-corrected Hβ mass with a separately recorded 0.5 dex virial calibration scale [14]. COSMOS3D-13852 also uses Hβ, but its source discusses possible 1–2 dex overestimates from scattering [38]. Applying those directional offsets to this object's point mass alone gives reference requirements of 1.115 and 0.916, respectively. These are illustrative source-motivated perturbations, not inferred corrections or a probability distribution. The common 0.5 dex experiment does not bracket that caveat.

Conversely, J0910_2028_12910 has a quoted 0.01 dex mass error and a reference requirement of 1.053. Its sampled probability exceeds 0.999, but the separately recorded 0.5 dex mass scale is far larger than the formal error [19]. It drops below unity in the negative-offset experiment. ZS7 illustrates a third error meaning: its quoted 0.4 dex already includes calibration scatter [18]. These objects cannot be ranked by quoted-error probability as though the errors expressed identical physical uncertainty.

5.3 Contrasting object-level constraints

UNCOVER-20466 and GS-20057765 illustrate strong primary-sample constraints on the reference history: their required averages are 1.452 and 1.355, respectively, with reported-error 16th percentiles above unity. Increasing the seed to $10^3 M_\odot$ leaves their point requirements above unity (1.217 and 1.101), whereas $10^5 M_\odot$ seeds reduce them to 0.746 and 0.592. At the lower efficiency of 0.05719, the corresponding light-seed requirements are 0.793 and 0.740. Their masses constrain combinations of seed mass and sustained growth; neither object uniquely selects a seeding channel. For a quiet state with zero accretion and an Eddington-limited active state, these latter averages would require active fractions of about 79 and 74 per cent over the entire available interval. Falling below unity therefore need not imply an easily sustained fuel supply.

GN-z11 illustrates a different kind of constraint. Its high redshift yields a reference point requirement of 1.437 despite its smaller adopted mass. Delaying seeding to $z = 20$ raises that requirement to 1.880,

Table 6: Source-linked interpretation of the complete primary threshold set. Caveats summarize the source records cited in Table 1; proposed observations are our assessment of the measurement that would most directly test the stated limitation, not scheduled or guaranteed feasible programmes.

Object	Mass-estimator caveat	Proposed observation
UNCOVER-20466	Hbeta; demagnified mass; lens model and BLR assumptions remain.	Independent lens model and broad-line profile constraints.
GS-20057765	Hbeta; quoted errors; extinction and calibration not fully quantified.	Deeper line profiles and an extinction constraint.
COSMOS3D-13852	Hbeta; source warns of possible 1–2 dex scattering bias.	Separate scattering wings from virial broadening.
RUBIES-EGS-55604	Hbeta; dust-corrected mass; separate 0.5 dex calibration scale.	Test dust correction and broad-line kinematics.
CEERS-1019	Hbeta; broad component has comparatively low significance.	Confirm the broad component with deeper spectroscopy.
GS-20030333	Hbeta; broad quoted errors; extinction/calibration caveats.	Improve line profile and extinction constraints.
GS-164055	Hbeta; reported interval straddles the reference threshold.	Reduce mass uncertainty with deeper spectroscopy.
GN-38509	Halpha; separate 0.3 dex calibration scale.	Independent estimator or calibration constraint.
J0910_2028_12910	Halpha; 0.01 dex formal error excludes 0.5 dex mass scale.	Independent mass calibration; smaller fit errors alone are insufficient.
CEERS-00717	Halpha; local calibration; no numeric systematic supplied.	Cross-check the virial estimator and line decomposition.
ZS7	Hbeta; quoted 0.4 dex already includes calibration scatter.	Spatially separate nuclear kinematics and interacting components.
GN-4685	Hbeta; reported interval straddles the reference threshold.	Improve line profile and extinction constraints.

exceeding UNCOVER-20466's 1.733 in that scenario. Thus even the identity of the most demanding object depends on the starting time. GN-z11's UV estimator excludes it from the primary sample; its probable evidence classification is permitted by the primary evidence cut. An independent, method-comparable mass would determine whether this apparent timing constraint survives the measurement assumptions. These examples distinguish sensitivity to the available growth time from sensitivity to mass methodology.

5.4 Limitations

The dominant limitations are scientific rather than computational. The catalogue is a union of heterogeneous selections and cannot support pooled demographics without source-specific completeness models. Single-epoch virial masses share calibration and geometry uncertainties that are incompletely reported across papers. The fixed-offset experiment quantifies one coherent mass-scale perturbation; it does not model source-specific biases, correlated calibration errors, or selection-dependent scatter. Two source-supported duplicate pairs have been reconciled without removing measurements. Three identity groups remain open: Baccus GDS_1210_9515 versus JADES GS-8083, GS-10013704 versus Scholtz 99671, and Scholtz 16745 versus 208643. The first requires target/spectral and preferred-mass reconciliation; the latter two require imaging and aperture checks. Unique-object and host counts remain provisional. All six affected records are excluded from manuscript inference; the inclusion/exclusion sensitivity above shows the effect on the reported results. This addresses their influence on the present analysis without claiming physical identity resolution. Future readmission requires source-backed spectral or imaging decisions, not wider matching thresholds. Finally, Eq. 2 compresses time-dependent accretion, feedback, mergers, and radiative-state changes into effective parameters. Its outputs should be read as conditional comparisons, not reconstructed histories. Reported-error probabilities condition on fixed growth parameters; marginalization requires justified parameter distributions and correlations, not just scenario maps. They are not seed-origin probabilities. The age relation neglects radiation; the supplementary $z_{\mathrm{seed}} = 3400$ plot is an extrapolation, not evidence for primordial formation or a physical radiation-era accretion history.

All admitted numerical mass/error fields have independent source checks; all 350 central redshifts and 323 available coordinate pairs are also checked against independent source evidence. Twenty-seven measurement rows lack coordinates. Redshift uncertainties, full source posteriors, and auxiliary observables are not exhaustively validated. Resolving the three groups is required before readmitting their records or claiming a final unique-object census. The literature cutoff is 3 September 2026; catalogue completeness refers to the declared source membership, not every JWST discovery. The external A2744-QSO1 comparison in Section 4.6 evaluates a pre-cutoff dynamical measurement without silently changing frozen membership.

6 Conclusions

The atlas provides a reproducible, source-linked comparison of early black-hole growth requirements. Its principal results are:

- The conservative 224-object primary sample contains 12 reference point requirements above unity and six with conditional $P \geq 0.95$ under quoted errors.
- UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604 retain the probability threshold after a -0.5 dex mass shift. Their distinct lensing, scattering, and calibration caveats identify different observational tests.
- Seed formation time changes both threshold membership and the most demanding object; lowering the fixed efficiency to 0.05719 removes all point requirements above unity. These are constraints on combinations of assumptions.
- The external A2744-QSO1 dynamical estimate increases its reference requirement from 0.929 to approximately unity, with an error interval spanning the threshold. Independent mass information can change the growth interpretation even without adding a new object.

The threshold results survive exclusion of the known ambiguous identities; the full catalogue's unique-object census remains provisional. The useful outcome is a set of named observational tests with explicit assumptions, not a universal ordering or a measurement of formation-channel frequencies.

Appendix A Source-family inventory

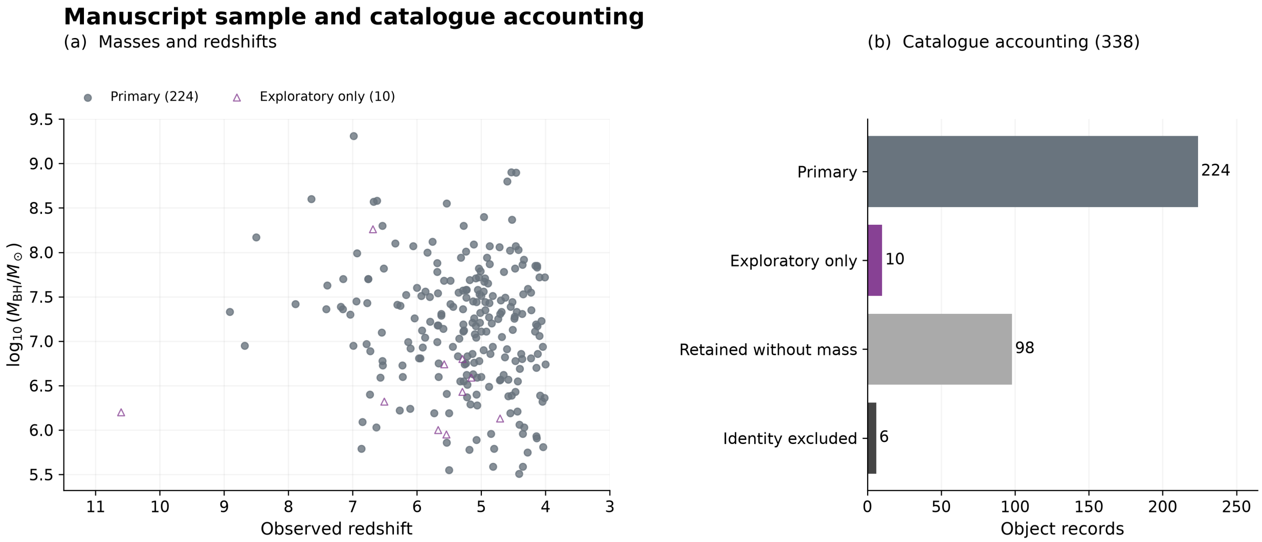

Figure 6: Manuscript selection. Left: 224 primary objects (circles) and ten exploratory-only objects (triangles). Right: mutually exclusive record counts, including 98 retained without masses and six identity exclusions. The exploratory numerical sample contains the primary sample and has 234 objects. Redshift decreases from left to right, corresponding to increasing cosmic time.

Table 7: v3 scale and analysis subsets.

Product or subset	Rows or objects
Measurements / provisional objects / hosts	350 / 338 / 337
Growth-eligible measurements / objects	244 / 237
Method-eligible primary measurements / objects	234 / 227
Manuscript primary / exploratory objects	224 / 234
Identity-excluded object records	6
Catalogue-only object records	101

The full catalogue contains 247 broad-line AGN, 49 narrow-line candidates, 30 photometric candidates, ten high-ionization candidates, and two X-ray candidates. Preferred-row evidence labels comprise 214 secure, 41 probable, 82 candidate, and one disputed classification. These full-catalogue counts precede the manuscript exclusions.

Table 8 gives every admitted source family. Counts distinguish measurements from preferred objects and avoid attributing one shared physical object independently to multiple sources. Auxiliary coordinates are attributed in the registry to THRILS [11], UHZ1 spectroscopy [22], and the versioned JADES DR3 GOODS-S prism catalogue. The machine-readable companion retains full selection-channel and caveat tags alongside source URLs. “Context” and “Limit” denote values retained outside numerical growth inference.

Table 8: Source-family inventory. Counts are admitted measurements / eligible mass measurements / preferred primary objects after identity exclusion (M/G/P). The columns sum to 350/244/224; G includes alternatives, whereas P counts each selected object once. "Balmer" denotes single-epoch virial masses.

Source	M/G/P	Mass	Selection channel; principal caveat
Baccus [19]	49/49/48	Balmer	NIRSpec broad lines; Frozen preprint masses; calibration
Chavez Ortiz [33]	1/0/0	Context	UV/optical diagnostics; Model-dependent mass
Chisholm [26]	1/0/0	None	High-ionization lines; No canonical mass
Davis / THRILS [10]	6/6/6	Balmer	EELG-selected spectroscopy; Virial calibration
Fei / GLIMPSE [20]	10/10/8	Balmer	Lensed broad lines; Lensing corrections
Greene / UNCOVER [13]	9/9/9	Balmer	Lensed broad lines; Lens model; BLR interpretation
Harikane [9]	10/10/5	Balmer	NIRSpec broad lines; Local virial calibration
Juodzbalis / JADES [5]	23/23/21	Balmer	Broad Balmer lines; Extinction; calibration
Killi [17]	1/1/1	Balmer	Broad Halpha; Linear mass-error conversion
Kocevski / LRDs [14]	6/6/6	Balmer	Broad Balmer lines; Dust correction; calibration
Larson / CEERS-1019 [16]	1/1/1	Balmer	Broad Hbeta; Broad-component significance
Lin / ASPIRE [8]	16/16/16	Balmer	Broad Halpha; Virial calibration
Lin / COSMOS-3D [38]	13/13/13	Balmer	NIRCam grism broad lines; Possible scattering bias
Lyu / SMILES [30]	19/0/0	None	MIRI SED selection; SED-dependent AGN identification
Maiolino / GN-z11 [25]	1/1/0	UV	UV lines; UV estimator outside primary
Mascia [34]	8/0/0	None	Compact blue broad emitters; No canonical mass
Matthee / EIGER-FRESCO [7]	20/20/19	Balmer	Broad Halpha; Dust; virial calibration
Mazzolari [28]	25/0/0	None	Narrow-line diagnostics; Star-formation overlap
Leung / MEOW [29]	12/0/0	None	MIRI SED selection; SED-dependent AGN fraction
Naidu / MoM-BH*-1 [36]	1/0/0	Context	Broad Hbeta / SED; Scattering; stellar alternative
Napolitano / GHZ9 [31]	1/0/0	Context	UV lines / X-ray evidence; Mass assumes Eddington rate
Napolitano / GHZ4,7 [39]	2/0/0	None	Narrow / high-ionization lines; Tentative diagnostics
Ren / ALPINE-CRISTAL [12]	7/7/1	Balmer	Host-selected broad Halpha; Conditional BLR interpretations
Scholtz / JADES [24]	21/0/0	None	Narrow / high-ionization lines; Tentative classifications
Skyfire / CEERS [15]	22/22/22	Balmer	Broad Halpha; Virial calibration
Tang [27]	3/0/0	None	High-ionization lines; Ionizing-source ambiguity
Taylor / CEERS-RUBIES [6]	37/37/35	Balmer	Broad Halpha; Slit loss; dust; calibration
Treiber / UNCOVER [35]	2/0/0	None	UV-line diagnostics; Stellar alternatives
Ubler / ZS7 [18]	1/1/1	Balmer	Offset broad Hbeta; Error includes calibration scatter
Bogdan / Zou / UHZ1 [21, 23]	2/0/0	Context	X-rays / SED; Later detection disputed
Zhang / LRDs [32]	5/0/0	Limit	Narrow lines / SED; Mass upper limits
Zhuang / NEXUS [37]	15/12/12	Balmer	NIRCam WFSS broad lines; Twelve masses lack errors

A.1 Source-admission checks

The v3 additions include 15 NEXUS measurements from NIRCam/WFSS [37], 13 COSMOS-3D NIRCam-grism measurements [38], and GHZ4/GHZ7 [39]. These contribute 30 measurements and 29 object records. NX10835 lies 0.066 arcsec from an existing Mascia identifier at consistent redshift, so the new mass-bearing measurement becomes preferred for the same object.

The Scholtz et al. source audit provides another cardinality check. The paper reports 42 objects, whereas its source-native table contains 41 rows. Exactly 21 table rows satisfy $z \geq 4$; all are admitted. One is identity-linked to a broad-line object, yielding 20 distinct narrow-line candidates from that source. JADES-NS-GS00099671 is retained with its published host and bolometric observables but without an invented black-hole mass.

A.2 Preferred measurements and identity exclusions

Stable measurement, object and host identifiers retain extraction locations, source versions and uncertainty semantics. Preferred measurements are chosen from source assessments, not inferred growth pressure or publication date alone. The CEERS-2782 group uses RUBIES-EGS-50052 because the source prefers its higher-signal-to-noise spectrum over the slit-loss measurement. NX10835 uses the newly available NEXUS mass; UHZ1 uses the later full-dataset evidence assessment and remains outside numerical growth inference. The measurement–object link table records each choice and its reason.

Before the conservative identity exclusions, there are 244 eligible measurements and 237 preferred objects: 236 Balmer-line and one UV single-epoch virial masses. The method-defined primary subset has 234 measurements and 227 objects. The six excluded records link to seven admitted measurements; the three with eligible masses are GDS_1210_9515, GS-8083 and GS-10013704. All three otherwise pass the primary cuts. The other three excluded records lack eligible masses. Exclusion does not establish duplication or resolve the identities. Seven of the nine additional candidate objects have conditional broad-line interpretations. All original measurements and catalogue flags remain available.

Appendix B Catalogue navigation score

The point navigation score is

$$S = \min\left[100, 100\max\left(C\left(\frac{f_2 - 0.3}{1.2}\right), C\left(\frac{s_{0.3} - 4}{2.8}\right)\right) + 8C\left(\frac{z-6}{4}\right)\right], \tag{4}$$

where $C(x) = \min(1, \max(0, x))$, f_2 is the required $f_{\rm Edd}$ for a $10^2\,M_\odot$ seed, and $s_{0.3}$ is the required $\log_{10}(M_{\rm seed}/M_\odot)$ at $f_{\rm Edd} = 0.3$. This heuristic combines two growth diagnostics and a redshift term; it is not a probability. Ties use decreasing f_2, decreasing redshift, then stable identifier. The manuscript's required-$f_{\rm Edd}$ table is instead ordered directly by f_2.

Appendix C Efficiency prescription for compatibility scenarios

For the illustrative spin-dependent maps, the ideal Kerr thin-disk efficiency is the ISCO binding efficiency [2]:

$$\epsilon_a = 1 - \sqrt{1 - \frac{2}{3r_{\rm ISCO}}}, \tag{5}$$

$$r_{\rm ISCO} = 3 + Z_2 - \mathrm{sgn}(a)\sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}, \tag{6}$$

$$Z_1 = 1 + (1 - a^2)^{1/3}\left[(1 + a)^{1/3} + (1 - a)^{1/3}\right], \qquad Z_2 = \sqrt{3a^2 + Z_1^2}. \tag{7}$$

The dimensionless spin is $a = cJ/(GM_{\rm BH}^2)$, where J is the signed angular momentum relative to the disk, and $r_{\rm ISCO}$ is in units of $GM_{\rm BH}/c^2$. The ideal efficiencies at $a = -1, 0, +1$ are approximately 0.038, 0.057, and 0.423, respectively, neglecting photon capture and stress at the ISCO. Logarithmic luminosity growth at supercritical inflow is motivated by advection and wind models [3]. Our unit-coefficient relation is a phenomenological ansatz, not a fit to those models or a treatment of radial mass loss. Above unity, the maps adopt

$$f_{\rm Edd} = 1 + \ln \dot{m}, \qquad \dot{m} = \frac{\dot{M}_{\rm in}}{L_{\rm Edd}/(\epsilon_a c^2)}, \qquad \epsilon_{\rm eff} = \begin{cases} \epsilon_a, & f_{\rm Edd} \leq 1, \\ \epsilon_a f_{\rm Edd} \exp(1 - f_{\rm Edd}), & f_{\rm Edd} > 1. \end{cases} \tag{8}$$

The logarithmic luminosity ansatz applies only on the super-Eddington branch. Each compatibility cell holds both the rate and its effective efficiency constant over the interval. Thus the cell cannot represent a variable history merely by substituting its average luminosity ratio. This is an illustrative parameter-scan prescription, not a relativistic slim-disk or spin-evolution calculation. If efficiency varies with time, the integrated growth depends on $\int [(1-\epsilon(t))/\epsilon(t)] f_{\rm Edd}(t)\, dt$; it cannot in general be represented by separately averaged efficiency and rate.

Figure 4 gives the full scenario grid. The broad 10^2–$10^6 M_\odot$ interval retains the legacy code key `pbh_origin_hypothesis` solely for artifact compatibility; it models no primordial formation process. The descriptive fractions use point masses and the inclusive interval criterion in Section 3.3, without error integration.

Appendix D Growth-model derivations and implementation

D.1 Accretion and exponential growth

The growth constraints follow from the relation between accretion luminosity, retained mass, and available cosmic time. For electron-scattering opacity in ionized hydrogen, the Eddington luminosity and luminosity-based Eddington ratio are

$$L_{\rm Edd} = \frac{4\pi G M_{\rm BH} m_p c}{\sigma_T}, \qquad f_{\rm Edd} = \frac{L_{\rm bol}}{L_{\rm Edd}}, \qquad t_{\rm Edd} = \frac{M_{\rm BH} c^2}{L_{\rm Edd}} = \frac{c\sigma_T}{4\pi G m_p} \simeq 0.45\ {\rm Gyr}. \tag{9}$$

Here G, c, m_p, and σ_T denote the gravitational constant, speed of light, proton mass, and Thomson cross section. With radiative efficiency $0 < \epsilon < 1$ and rest-mass inflow rate $\dot{M}_{\rm in}$,

$$L_{\rm bol} = \epsilon \dot{M}_{\rm in} c^2, \qquad \dot{M}_{\rm BH} = (1-\epsilon)\dot{M}_{\rm in} = \frac{1-\epsilon}{\epsilon}\frac{f_{\rm Edd}}{t_{\rm Edd}} M_{\rm BH}. \tag{10}$$

The inflow rate is defined at the radiating flow; additional mass loss before that point is not modelled. Dividing Eq. 10 by $M_{\rm BH}$ and integrating at fixed ϵ gives

$$\ln\left(\frac{M_{\rm BH}}{B_{\rm merge} M_{\rm seed}}\right) = \frac{1-\epsilon}{\epsilon}\frac{1}{t_{\rm Edd}} \int_{t(z_{\rm seed})}^{t(z_{\rm obs})} f_{\rm Edd}(t)\, dt = \frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}}. \tag{11}$$

Exponentiating yields Eq. 2; solving for the average rate and applying the non-negative floor yields Eq. 3. For a constant positive $f_{\rm Edd}$, the e-folding time is

$$t_{\rm efold} = \frac{\epsilon}{1-\epsilon}\frac{t_{\rm Edd}}{f_{\rm Edd}}. \tag{12}$$

It is 0.05 Gyr at $\epsilon = 0.1$ and $f_{\rm Edd} = 1$.

D.2 Merger multiplier and inverse seed mass

The implementation evaluates Eq. 2 in log mass:

$$\log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm seed}}{M_\odot}\right) + \log_{10} B_{\rm merge} + \frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}\ln 10}. \tag{13}$$

We evaluate $B_{\rm merge} = 1$ (no boost) and 2 (a factor-of-two mass boost). At fixed accretion history,

$$\frac{M_{\rm BH}(B_{\rm merge}=2)}{M_{\rm BH}(B_{\rm merge}=1)} = 2, \qquad \Delta \log_{10} M_{\rm BH} = \log_{10} 2 \simeq 0.3010. \tag{14}$$

Solving the same relation for seed mass gives

$$\log_{10}\left(\frac{M_{\rm seed,req}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) - \log_{10} B_{\rm merge} - \frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}\ln 10}. \tag{15}$$

Here $M_{\rm seed,req}$ is the seed mass required for a specified average rate, efficiency, growth interval and merger multiplier.

D.3 Cosmic time

Cosmic ages use the flat matter-plus-Lambda relation

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \operatorname{asinh}\left[\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}(1+z)^{-3/2}\right], \tag{16}$$

with $H_0 = 67.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$. The Hubble constant is converted to inverse time; ages and growth intervals are in Gyr. Radiation is neglected. In particular, the supplementary $z_{\rm seed} = 3400$ comparison extrapolates this age relation and is not a radiation-era seed-formation calculation.

D.4 Two-state duty cycle

For a two-state history with the same efficiency in both states,

$$\overline{f_{\rm Edd}} = D f_{\rm burst} + (1-D) f_{\rm quiet}, \qquad D_{\rm req} = \frac{\overline{f_{\rm Edd}}_{\rm req} - f_{\rm quiet}}{f_{\rm burst} - f_{\rm quiet}}. \tag{17}$$

Here D is the active-state time fraction, $f_{\rm burst} > f_{\rm quiet} \geq 0$, and the diagnostic requires $\overline{f_{\rm Edd}}_{\rm req} \geq f_{\rm quiet}$. Values $D_{\rm req} > 1$ identify an infeasible specified two-state history and are retained. An average growth requirement is not an instantaneous Eddington ratio or a uniquely measured duty cycle.

D.5 Sampling, mass conversions and track selection

Draws use seed 20260808 and a stable identifier-based random stream. For 10,000 draws, the binomial Monte Carlo standard error of a probability is at most 0.005 (approximately 0.0022 at probability 0.95). This is numerical sampling error, separate from astrophysical uncertainty. The twelve mass-bearing NEXUS objects have no published mass errors: repeated point values produce zero-width stored intervals, while probabilities and uncertainty ranks remain unavailable. Source errors are retained as quoted and need not be purely statistical. For ZS7, the published 0.4-dex error already includes virial-calibration scatter. Separately reported method systematics are stored and, where possible, mapped to sensitivity envelopes without assuming independent Gaussian contributions.

The mass-scale tests translate each object's original draws by the fixed offset; no scatter is added in quadrature and no marginalization over the offset is performed. The final published Killi mass $8^{+0.5}_{-0.4} \times 10^8 M_\odot$ is converted by transforming both linear bounds, giving $\log_{10}(M/M_\odot) = 8.90309^{+0.02633}_{-0.02228}$.

For Figure 1, candidate average rates span 0.1–2.0 in steps of 0.1. For each seed mass and fixed efficiency, we retain the two rates with the most primary objects within 0.5 dex of either the $B = 1$ or $B = 2$ prediction, requiring at least five such objects. Ties favour smaller median absolute residuals, then smaller rates. Objects at $z \simeq 4$–6 dominate this display selection; the tolerance is not an uncertainty interval. Baseline rankings, duty-cycle diagnostics and seed-redshift–mass maps retain $\epsilon = 0.1$. The heuristic catalogue score in Appendix B is separate from the required-rate ranking used for scientific conclusions.

Appendix E Data and code availability

The catalogue, source registries, code, and manuscript analysis are available at `https://github.com/nniickels/highz-accretion-atlas`. The frozen v3 catalogue baseline is identified by Git commit `a40a0d2`; version manifests under `releases/` record the dataset artifact hashes. The revised manuscript's source-linked comparisons and generated tables are in `paper/analysis/`, with explicit external inputs in `paper/review_inputs.json`. A permanent archival DOI has not yet been assigned; the repository and Git baseline provide the current access and version identification.

Five ordered notebooks reproduce the catalogue, science products, figures, and atlas using pinned dependencies. Reproduction checks compare regenerated numerical products, figures, and manuscript table fragments with an independent stored baseline, then verify their internal consistency. Independent cosmic-age quadrature and growth integration supplement the regression checks. The source-family inventory in Appendix A links measurements, methods, selection channels, and major caveats. Full identifiers, source-native observables, alternative measurements, and per-object maps remain available.

Source agreement does not establish estimator validity or astrophysical uniqueness. The manuscript exclusion check verifies that all open identity groups are withheld; the separate full-catalogue identity gate remains open for the three groups listed in Section 5.4.

References

[1] Shen, Y. 2013, The Mass of Quasars, arXiv:1302.2643v1.

[2] Bardeen, J. M., Press, W. H., & Teukolsky, S. A. 1972, *ApJ*, 178, 347–369, doi:10.1086/151796.

[3] Poutanen, J., Lipunova, G., Fabrika, S., Butkevich, A. G., & Abolmasov, P. 2007, *MNRAS*, 377, 1187–1194, doi:10.1111/j.1365-2966.2007.11668.x.

[4] Dayal, P. 2024, *A&A*, 690, A182, doi:10.1051/0004-6361/202451481.

[5] Juodžbalis, I. et al. 2026, *MNRAS*, 546, stag086, doi:10.1093/mnras/stag086.

[6] Taylor, A. J. et al. 2025, *ApJ*, 986, 165, doi:10.3847/1538-4357/add15b.

[7] Matthee, J. et al. 2024, *ApJ*, 963, 129, doi:10.3847/1538-4357/ad2345.

[8] Lin, X. et al. 2024, *ApJ*, 974, 147, doi:10.3847/1538-4357/ad6565.

[9] Harikane, Y. et al. 2023, *ApJ*, 959, 39, doi:10.3847/1538-4357/ad029e.

[10] Davis, K. et al. 2026, arXiv:2602.23310v1.

[11] Hutchison, T. A. et al. 2025, THRILS redshift catalogue, arXiv:2512.12509v1.

[12] Ren, J. et al. 2025, *MNRAS*, 544, 211–233, doi:10.1093/mnras/staf1709.

[13] Greene, J. E. et al. 2024, *ApJ*, 964, 39, doi:10.3847/1538-4357/ad1e5f.

[14] Kocevski, D. D. et al. 2025, *ApJ*, 986, 126, doi:10.3847/1538-4357/adbc7d.

[15] Skyfire Collaboration 2026, arXiv:2609.00112v1.

[16] Larson, R. L. et al. 2023, *ApJL*, 953, L29, doi:10.3847/2041-8213/ace619.

[17] Killi, M. et al. 2024, *A&A*, 691, A52, doi:10.1051/0004-6361/202348857.

[18] Übler, H. et al. 2024, *MNRAS*, 531, 355, doi:10.1093/mnras/stae943.

[19] Baccus, C. et al. 2026, *ApJ*, 1006, 165, doi:10.3847/1538-4357/ae7de7.

[20] Fei, Q. et al. 2026, *ApJ*, 1003, 244, doi:10.3847/1538-4357/ae6248.

[21] Bogdán, Á. et al. 2024, *Nature Astronomy*, 8, 126, doi:10.1038/s41550-023-02111-9.

[22] Goulding, A. D. et al. 2023, *ApJL*, 955, L24, doi:10.3847/2041-8213/acf7c5.

[23] Zou, F. et al. 2026, UHZ1 full-data X-ray reanalysis, accepted by *ApJ*, arXiv:2603.24893v2.

[24] Scholtz, J. et al. 2025, *A&A*, 697, A175, doi:10.1051/0004-6361/202348804.

[25] Maiolino, R. et al. 2024, *Nature*, 627, 59–63, doi:10.1038/s41586-024-07052-5.

[26] Chisholm, J. et al. 2024, *MNRAS*, doi:10.1093/mnras/stae2199.

[27] Tang, M. et al. 2025, accepted by *ApJ*, arXiv:2505.06359v2.

[28] Mazzolari, G. et al. 2024, *A&A*, 691, A338, doi:10.1051/0004-6361/202451860.

[29] Leung, G. C. K. et al. 2026, submitted to *ApJ*, arXiv:2607.02666v1.

[30] Lyu, J. et al. 2024, *ApJ*, 966, 229, doi:10.3847/1538-4357/ad3643.

[31] Napolitano, L. et al. 2025, *ApJ*, 989, 75, doi:10.3847/1538-4357/ade706.

[32] Zhang, Z. et al. 2026, accepted by *ApJ*, arXiv:2506.04350v2.

[33] Chavez Ortiz, O. A. et al. 2025, revised 2026, arXiv:2511.03035v2.

[34] Mascia, S. et al. 2026, submitted to *A&A*, arXiv:2608.25021v1.

[35] Treiber, H. et al. 2025, *ApJ*, 984, 93, doi:10.3847/1538-4357/adc38f.

[36] Naidu, R. P. et al. 2026, *Nature*, 656, 329–333, doi:10.1038/s41586-026-10846-4.

[37] Zhuang, M.-Y. et al. 2025, NEXUS broad-line AGN sample, arXiv:2505.20393v1.

[38] Lin, X. et al. 2025, COSMOS-3D broad-line AGN sample, accepted by *ApJ*, doi:10.3847/1538-4357/ae1b9b.

[39] Napolitano, L. et al. 2025, *A&A*, 693, A50, doi:10.1051/0004-6361/202452090.

[40] Juodžbalis, I. et al. 2025, A direct black hole mass measurement in a Little Red Dot at the Epoch of Reionization, arXiv:2508.21748v2; related publication doi:10.1038/s41586-026-10579-4.